Narxoz University
School of Digital Technologies

INDIVIDUAL ASSIGNMENT
ANALYSIS OF RETURNS AND RISK FOR A SELECTED STOCK
BYD Company Limited (002594.SZ)

Prepared by: Akmurzina Mariza
Instructor: Musulmankulova Aigul

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1. Company Description and Risk Identification

1.1. Company Overview

BYD Company Limited (ticker 002594.SZ, Shenzhen Stock Exchange) is a Chinese manufacturer of new energy vehicles (battery-electric and plug-in hybrid cars), batteries and electronic components. The main source of revenue is the sale of vehicles: about 80 % of sales come from the domestic market of China [1], and the rest comes from exports, which the company supports by building plants abroad, in particular in Hungary and Turkey [2]. Because BYD sells mainly in one country, its revenue is sensitive to Chinese consumer demand and to government policy.

The revenue of the company depends on several external factors. The first is consumer demand and regulation in China: the authorities are restricting the price war in the car industry, which reduced the competitive advantage of BYD [3]. The second is trade policy and geopolitics: high tariffs effectively close the US market, and the European Union and Mexico are considering limits on imports of cheaper Chinese vehicles [3]. The third is the exchange rate: the shares are quoted in CNY, export revenue is received in foreign currencies, and for an investor from Kazakhstan the result also depends on the CNY/KZT rate, which changed during the sample period (Section 2.6). The fourth is commodity prices, mainly lithium and other battery materials, which determine production costs.

1.2. Stock Data

For this analysis, BYD Company Limited shares listed on the Shenzhen Stock Exchange were selected.

Parameter	Description
Company	BYD Company Limited
Stock ticker	002594.SZ
Stock exchange	Shenzhen Stock Exchange
Currency	Chinese Yuan (CNY)
Data source	Yahoo Finance
Analysis period	October 19, 2022 – October 19, 2026
Data frequency	Daily
Price series	Adjusted Close
Number of observations	969

Historical stock prices were obtained from Yahoo Finance using the quantmod package in RStudio. Adjusted closing prices were selected because they account for relevant corporate actions, such as stock splits and dividends.

1.3. External Factors Affecting the Company

BYD's financial performance is influenced by several external factors.

Consumer demand. Demand for electric and hybrid vehicles directly influences the company's sales and revenue. Changes in consumer preferences and competition may affect vehicle prices and sales volumes.

Raw material prices. Battery production requires materials such as lithium, nickel, and other metals. Fluctuations in their prices may affect production costs and profit margins.

Government regulations. Environmental regulations, electric vehicle subsidies, import tariffs, and international trade policies can influence BYD's competitiveness in domestic and foreign markets.

Exchange rates. Currency fluctuations, particularly changes in the value of the Chinese yuan against foreign currencies, may influence international revenue, costs, and profitability.

1.4. Identification of Key Risks

Three major risks were identified for BYD Company Limited.

Risk	Classification	Description
Increasing competition	Market risk	Strong competition in the electric vehicle market may lead to lower selling prices and reduced profit margins.
Changes in government policies	Regulatory risk	Changes in subsidies, environmental regulations, and international trade tariffs may negatively affect sales and international expansion.
Currency fluctuations	Currency risk	Changes in exchange rates may affect revenue from international markets and the company's profitability.

References:

1. BYD Company Limited. *Company Profile and Business Overview*. https://www.byd.com/en/about-byd
2. BYD Company Limited. *Annual Report 2025*. https://static.cninfo.com.cn/finalpage/2026-03-31/1225052526.PDF
3. Yahoo Finance. *BYD Company Limited (002594.SZ) Historical Data*. https://finance.yahoo.com/quote/002594.SZ/history/

2. Data Preparation and Exploratory Data Analysis

2.1. Data Preparation

Historical daily stock prices of BYD Company Limited (002594.SZ) were obtained from Yahoo Finance using the quantmod package in RStudio.

The dataset covers the period from October 19, 2022, to October 19, 2026, and contains 969 daily price observations.

Adjusted closing prices were selected for the analysis because they account for corporate actions such as stock splits and dividends.

The dataset was checked for missing values and duplicate dates. No missing values or duplicate dates were detected in the original price data.

Daily logarithmic returns were calculated using the following formula: $r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$

- r_t - daily logarithmic return;
- ln - adjusted closing price on day (t);
- P_{t-1} - adjusted closing price on the previous trading day.

After calculating logarithmic returns, 968 valid return observations were obtained. One missing return was generated for the first trading day because the previous day's price was unavailable.

No zero returns were found in the dataset.

2.2. Stock Price Analysis

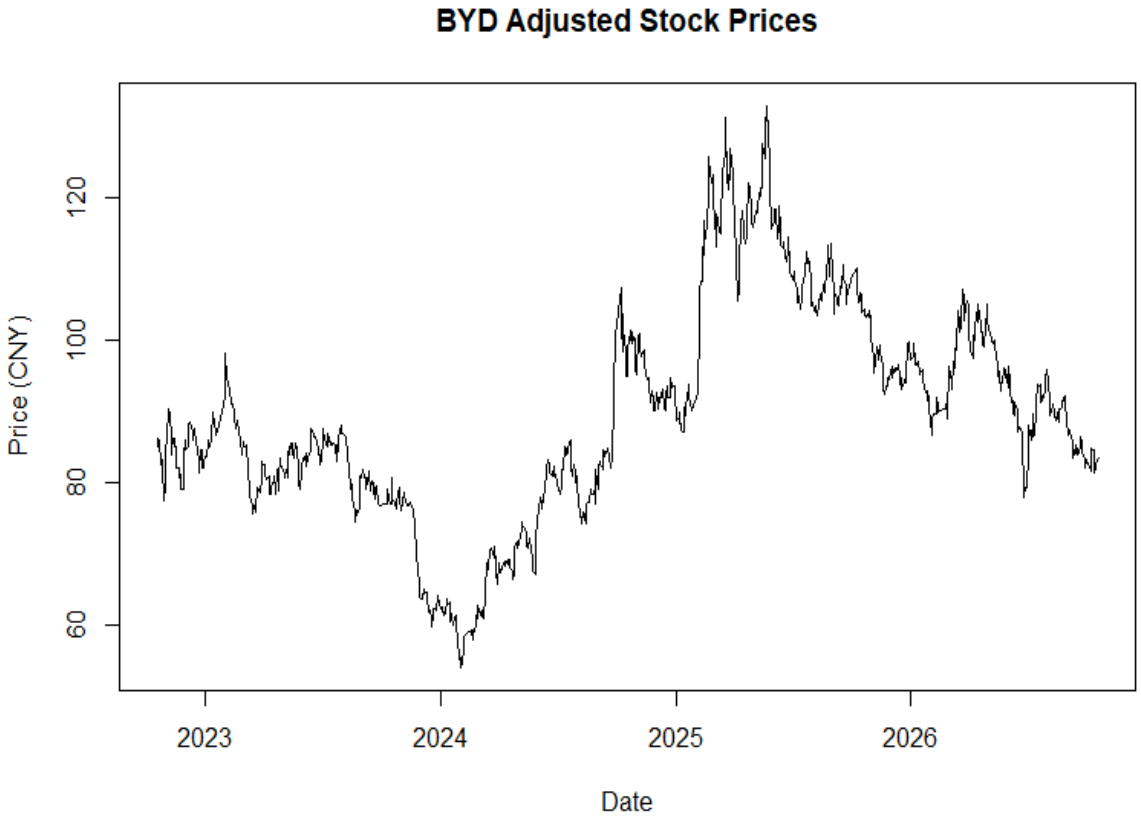


Figure 1. Adjusted closing prices of BYD Company Limited, 2022–2026 (CNY).

The adjusted price (Figure 1) went through several phases. In 2022–2023 it fluctuated between approximately 75 and 98 CNY, starting at 86,34 CNY on October 19, 2022. It then declined to the sample minimum of 54,14 CNY on February 2, 2024, and recovered strongly through 2024 and early 2025, reaching the sample maximum of 132,91 CNY on May 23, 2025. Since then the price has been declining, and on October 19, 2026 it was 83,50 CNY, close to its level at the beginning of the sample.

2.3. Daily Logarithmic Returns

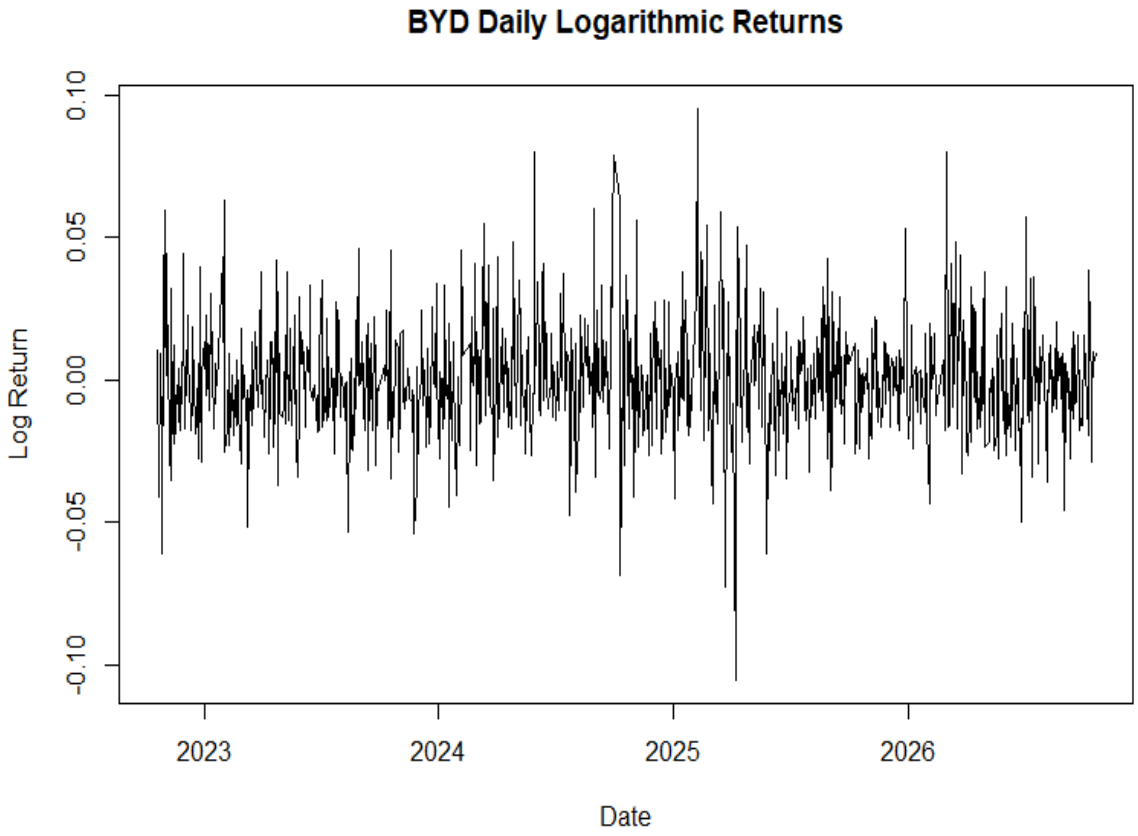


Figure 2. Daily logarithmic returns of BYD Company Limited, 2022–2026.

The returns (Figure 2) fluctuate around zero, but their volatility is not constant, and large movements appear in clusters. The first cluster is observed in the autumn of 2024: +7,89 % on 30.09.2024, +6,33 % on 08.10.2024 and −6,86 % on 09.10.2024 (logarithmic returns, in %). The second and most intense cluster covers February–April 2025: +9,53 % on 06.02.2025, −7,25 % on 21.03.2025 and the sample minimum of −10,54 % on 07.04.2025. Separate large movements were recorded on 29.05.2024 (+8,02 %) and on 02.03.2026 (+8,03 %). Among the ten largest absolute returns, six are positive and four are negative, so the risk is not limited to declines.

The sample maximum (+9,53 % logarithmic, which is +10,00 % simple return, on 06.02.2025) and the sample minimum (−10,54 % logarithmic, which is −10,00 % simple return, on 07.04.2025) coincide with the ±10 % daily price limit of the Shenzhen Stock Exchange, so the true unconstrained movements could have been larger.

2.4. Descriptive Statistics

Table 1. Descriptive statistics of BYD daily logarithmic returns

	Mean	Median	SD	Variance	Min	Max	Skewness	Kurtosis
1	-3e-05	-0.00153	0.01987	0.00039	-0.10536	0.09531	0.35573	5.50129

The mean daily return is practically zero (−0,00003 per day). It corresponds to a cumulative change of −3,29 % over the 968 days, which agrees with the price moving from 86,34 CNY to 83,50 CNY, so the stock gave no systematic profit over the period. The standard deviation of 0,01987 (1,99 % per day, or 31,54 % per year) shows a high level of risk. The median (−0,00153) is negative, which means that at least half of the trading days ended with a price decline; the mean is higher than the median by 0,00150 because of large positive returns, which is consistent with the positive skewness of 0,356 (the right tail is longer). At the same time, the largest movement in

absolute value is a loss (−0,10536 against +0,09531). The kurtosis of 5,501 (excess kurtosis 2,501) is well above 3, so extreme returns occur more often than the normal distribution predicts.

2.5. Distribution of Daily Returns

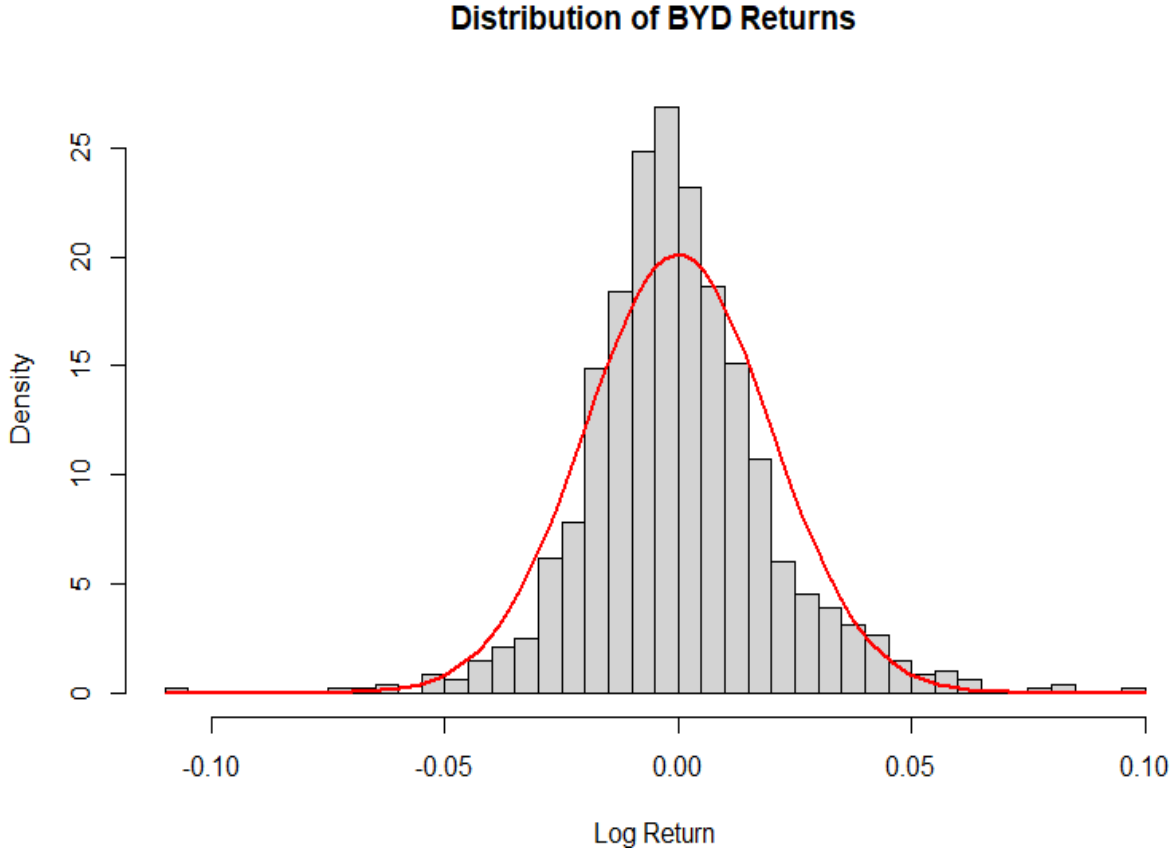


Figure 3. Histogram of BYD daily logarithmic returns with fitted normal density curve.

Figure 3 shows the empirical distribution of daily logarithmic returns, together with a normal density curve estimated using the sample mean and standard deviation. The histogram displays a pronounced central peak and observations extending into both tails. The empirical histogram and fitted normal curve do not coincide perfectly, especially near the central peak and in the tails. The higher empirical concentration of returns near zero, together with relatively extreme observations, suggests that the normal distribution may not fully capture the characteristics of BYD's daily returns. In particular, a normal model could underestimate the probability of unusually large price movements. Nevertheless, visual inspection alone is insufficient to reject normality. Formal goodness-of-fit tests and model comparisons are required to determine which distribution provides the most appropriate statistical description.

2.6. Five Largest Daily Losses

The five largest daily percentage declines were identified using simple daily percentage returns calculated from adjusted closing prices.

Table 2. Five largest daily losses of BYD shares

```
> largest_losses
          Date ReturnPct LossCNY
598 2025-04-07    -10.00   11.73
588 2025-03-21     -7.00    9.19
478 2024-10-09     -6.63    7.12
630 2025-05-26     -5.93    7.88
8   2022-10-28     -5.93    4.88
```

The causes of three of these losses were identified from news sources and are described below.

07.04.2025 (10,00 %). The Chinese market collapsed: the Shanghai Composite closed 7,34 % lower and the Shenzhen Component closed 9,66 % lower, and more than 2900 stocks fell to the daily limit [7, 8]. The sell-off followed the escalation of the US–China tariff war, including the retaliatory tariffs of 34 % announced by China [7]. The loss of exactly 10,00 % coincides with the daily price limit, so it was most likely a limit-down day.

09.10.2024 (6,63 %). Chinese stocks fell as the stimulus-driven rally faded: investors took profits after a strong rise, and the press conference of the National Development and Reform Commission did not give new details of stimulus. The Shanghai Composite fell by 6,6 % and the CSI 300 by 7,1 %, the largest daily losses since the beginning of the pandemic [9, 10].

26.05.2025 (5,93 %). BYD announced wide-ranging discounts on its models on the previous Friday, which raised fears of a new price war: its Hong Kong-listed shares fell by 8,6 % on that Monday, and the Shenzhen-listed shares also declined [11, 12].

For the losses on 21.03.2025 and 28.10.2022 a specific cause could not be reliably identified from available sources, so it is not stated. Four of the five largest losses occurred between October 2024 and May 2025, and the identified causes are external shocks (the trade war and expectations of state stimulus) and industry competition rather than company-specific financial failures.

2.7 Summary of the exploratory data analysis

The analysis of BYD shares (002594.SZ, Yahoo Finance) covered 969 adjusted daily closing prices from 19.10.2022 to 19.10.2026 and produced 968 logarithmic returns; no missing values, duplicates or zero returns were found. The price fell from 86,34 CNY to a minimum of 54,14 CNY on 02.02.2024, rose to a maximum of 132,91 CNY on 23.05.2025 and then declined to 83,50 CNY, with clusters of high volatility in the autumn of 2024 and in February–April 2025. The mean daily return was −0,00003, which is practically zero, while the standard deviation was 0,01987 (1,99 % per day) and the median was negative (−0,00153). The skewness of 0,356 and the kurtosis of 5,501, which exceeds the normal value of 3, show that the distribution is asymmetric and has heavier tails than the normal one. The histogram confirms a pronounced central peak and extreme values that the normal curve does not capture. The largest daily loss was 10,00 % (7,00 % and 6,63 % in the second and third place) on 07.04.2025, and the largest gain was +10,00 % on 06.02.2025; both coincide with the daily price limit. Therefore, extreme price movements are an important component of the investment risk of BYD, and the next section tests which probability distributions describe returns, losses and loss frequency better.

3. Fitting Probability Distributions

3.1. Distribution of Daily Returns

To analyze the statistical characteristics of BYD's daily stock returns, three probability distributions were fitted: Normal, Student's t, and Skewed Student's t. These models were selected because they allow different assumptions about the shape, symmetry, and tail behavior of financial returns.

The analysis was based on 968 daily logarithmic return observations from October 19, 2022, to October 19, 2026. All model parameters were estimated using the Maximum Likelihood Estimation (MLE) method.

The Normal distribution was used as the baseline model. It assumes that returns are symmetrically distributed around their mean and that extreme changes occur relatively infrequently.

Student's t distribution was introduced to account for heavier tails, which are frequently observed in financial data. Unlike the Normal distribution, Student's t assigns higher probabilities to extreme positive and negative returns.

Finally, the Skewed Student's t distribution was fitted to account for both heavy tails and asymmetry. This model is more flexible because it allows extreme negative and positive returns to follow different patterns.

Table 5. Estimated parameters of return distributions

Model	**Parameters**
Normal	$\mu = -0.000035$; $\sigma = 0.019863$
Student's t	$\mu = -0.001070$; $\sigma = 0.014756$; $\nu = 4.174$
Skewed Student's t	Four parameters: ξ, ω, α, ν; estimated by MLE

The Normal distribution produced an estimated mean close to zero and a standard deviation of approximately 0.01986, representing about 1.99 percentage points in daily logarithmic returns.

For Student's t, the estimated degrees of freedom were approximately 4.174. This relatively low value indicates heavier tails compared with the Normal distribution. Consequently, Student's t assigns greater probability to unusually large price movements.

The Skewed Student's t model additionally incorporates an asymmetry parameter, allowing the shape of the distribution to differ between positive and negative returns.

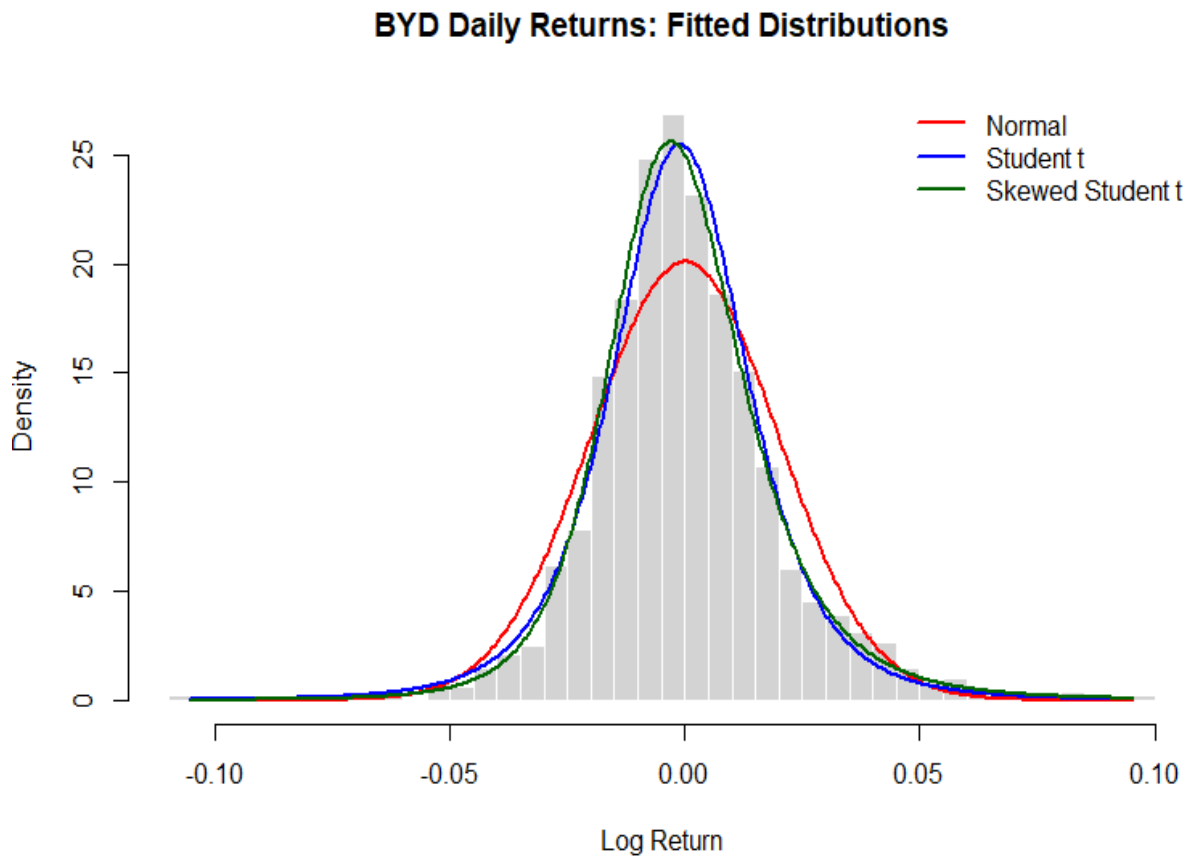


Figure 4. Histogram of BYD daily logarithmic returns with Normal, Student's t, and Skewed Student's t fitted density curves.

Figure 4 compares the observed return distribution with the three fitted probability density functions.

The histogram shows a pronounced concentration of returns around zero, together with relatively long tails on both sides. The Normal distribution, represented by the red curve, has a lower central peak and does not adequately represent the concentration of observations near zero.

Student's t, represented by the blue curve, follows the central peak more closely and provides greater flexibility in the tails. The Skewed Student's t distribution, represented by the green curve, offers an additional adjustment for asymmetry.

The fitted curves suggest that distributions with heavier tails are better suited to BYD's daily returns than the Normal distribution. This observation is consistent with the previously calculated kurtosis of 5.501, which exceeds the Normal distribution benchmark of 3.

A more detailed comparison using AIC, BIC, Q–Q plots, and goodness-of-fit tests is presented in Section 4.

3.2. Distribution of Loss Size

The second part of the analysis focuses exclusively on trading days when BYD shares experienced negative returns.

Loss size was defined as the absolute value of a negative daily logarithmic return expressed in percentage units: $X_t = -100r_t,\ r_t < 0$

where X_t represents the positive magnitude of a daily logarithmic loss.

Out of 968 valid daily return observations, 520 trading days recorded negative returns. Therefore, the proportion of loss days was:

Thus, negative returns occurred on approximately 53.72% of the observed trading days, while the remaining 46.28% were days with positive returns. No zero-return days were recorded.

Two probability distributions were fitted to the positive loss observations: Gamma and Lognormal.

The Gamma distribution was selected because it can represent positive, right-skewed variables with different degrees of variability. The Lognormal distribution was considered because it is commonly used to describe positive values with long right tails.

Table 4. Estimated parameters of loss size distributions

Model	Parameter	Estimate
Gamma	Shape (α)	1.253
Gamma	Rate (β)	0.923
Lognormal	Meanlog (μ)	−0.144
Lognormal	Sdlog (σ)	1.134

The Gamma distribution has a shape parameter of approximately 1.253 and a rate parameter of 0.923. These parameters indicate a positively skewed distribution in which relatively small losses are more common than large losses.

The Lognormal distribution was fitted using a meanlog of −0.144 and an sdlog of 1.134. Its relatively large logarithmic standard deviation produces a distribution with a long right tail.

Comparison of theoretical and empirical moments

The fitted models were compared with the sample mean and variance of observed loss sizes.

Table 5. Empirical and theoretical loss size statistics

Model	Mean	Variance
Empirical	1.35711	1.45603
Gamma	1.35705	1.46948
Lognormal	1.64729	7.10590

The empirical mean loss size was approximately 1.357%, with a variance of 1.456 percentage points squared.

The Gamma model produced a theoretical mean of 1.357%, almost identical to the empirical value. Its theoretical variance of 1.469 was also close to the observed variance.

In contrast, the Lognormal model produced a higher theoretical mean of 1.647% and a substantially larger variance of 7.106. This indicates that the fitted Lognormal distribution places considerably more weight on large losses than is supported by the overall sample moments.

The comparison suggests that the Gamma distribution provides a more representative description of typical loss sizes and their variability.

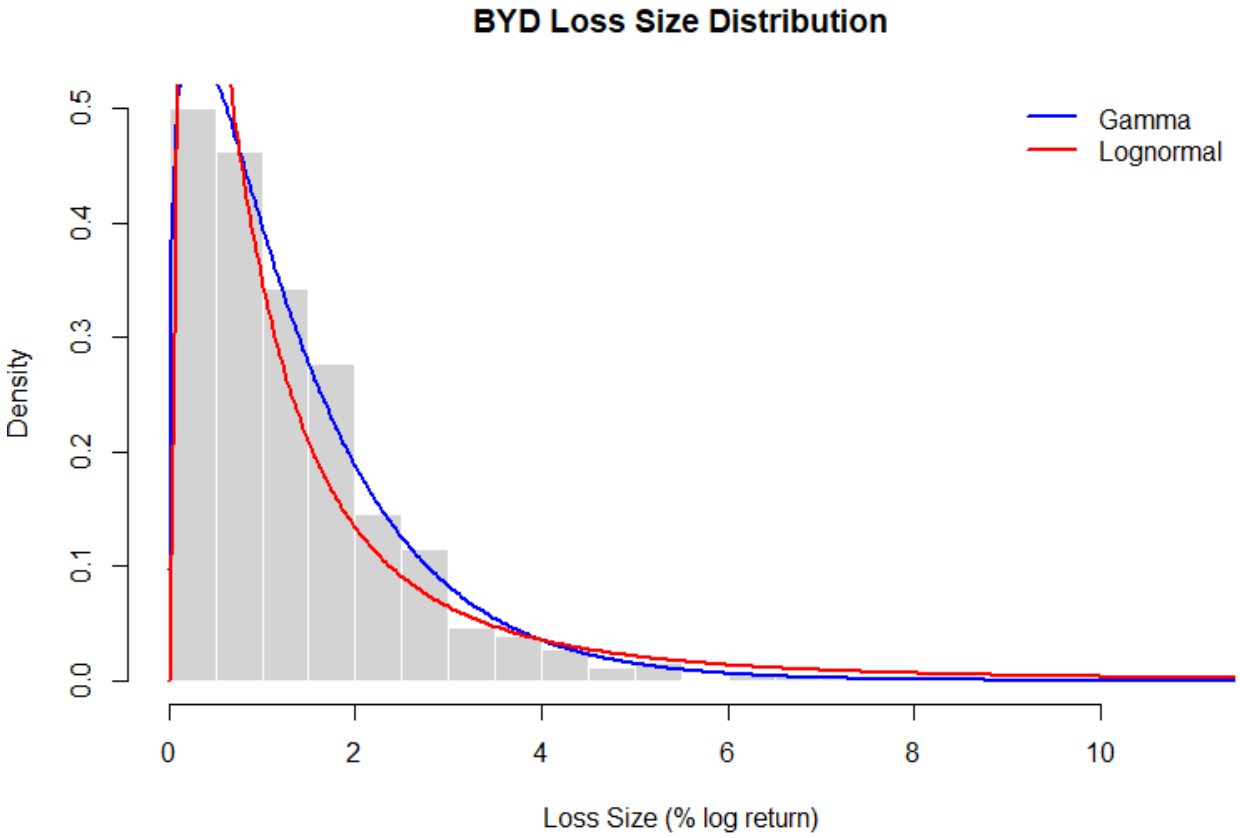


Figure 5. Histogram of positive BYD daily logarithmic losses with fitted Gamma and Lognormal density curves.

Figure 5 shows that most observed losses are relatively small, while large losses occur less frequently. The empirical distribution is positively skewed, with a long right tail extending toward the largest observed loss.

The Gamma density follows the main body of the histogram more closely, particularly in the range of small and moderate losses. The Lognormal distribution assigns greater probability to very small values and exhibits a longer tail.

Although the Gamma model provides a better visual fit over much of the observed range, differences remain in the extreme right tail. These differences are important because the probability of unusually large losses is a central concern in financial risk analysis.

Therefore, the distribution fitting results must be supplemented with formal goodness-of-fit tests before drawing final conclusions.

3.3. Distribution of Loss Frequency

The third variable analyzed was the monthly frequency of substantial stock price declines.

A substantial loss event was defined as a trading day when the simple daily percentage return was lower than −2%.

For each calendar month, the number of such events was calculated. In total, the analysis covered 49 calendar months, including partial months at the beginning and end of the observation period.

Two count distributions were compared: Poisson and Negative Binomial.

The Poisson distribution assumes that events occur with a stable average intensity and that the expected count is equal to its variance. The Negative Binomial distribution is more flexible because it allows the variance to exceed the mean.

Differences in the number of trading days across months were incorporated using an exposure adjustment. Specifically, the expected monthly event count was modeled as the daily event rate multiplied by the number of trading days in the corresponding month.

The estimated Poisson daily event rate was approximately 0.1105 events per trading day, corresponding to roughly 2.21 expected events over 20 trading days.

The Negative Binomial model estimated a very large dispersion parameter, approaching the Poisson limit. This suggests that introducing additional overdispersion did not substantially improve the description of monthly loss frequency.

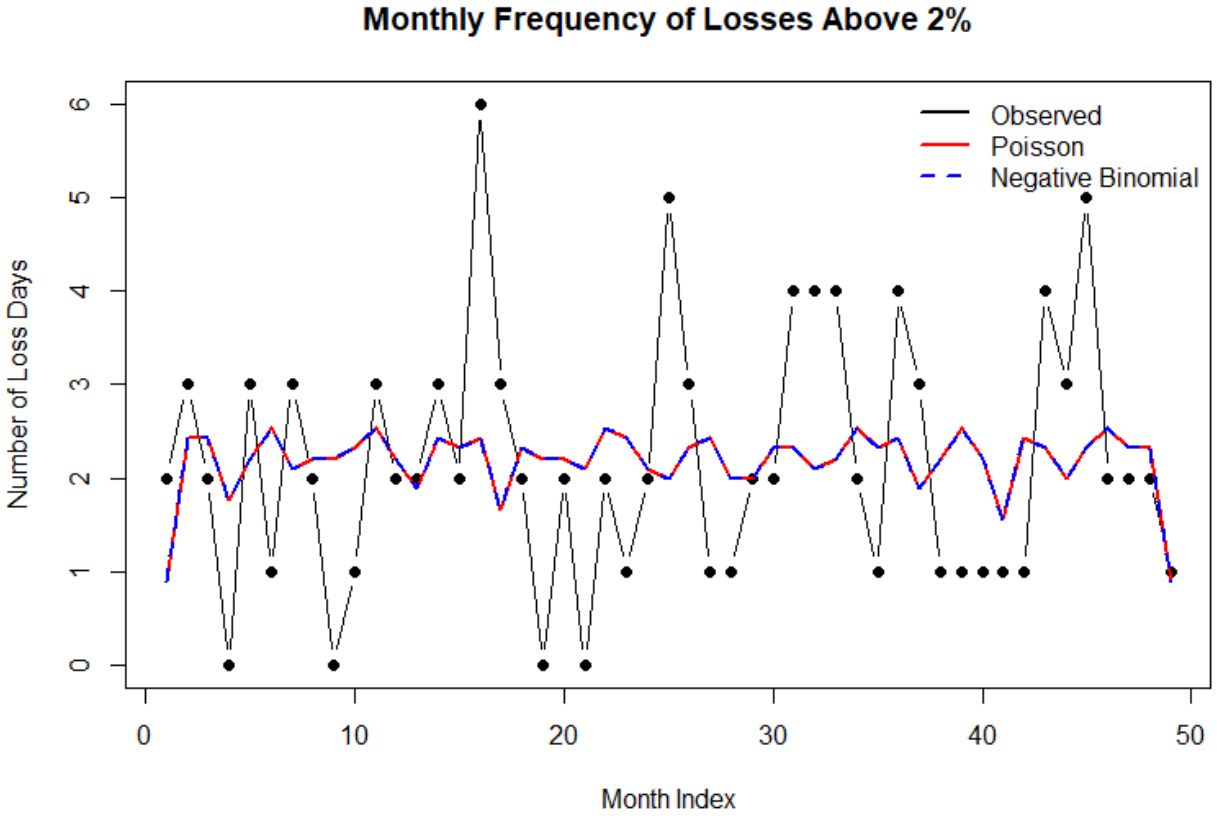


Figure 6. Observed monthly frequency of BYD stock declines exceeding 2%, compared with Poisson and Negative Binomial fitted expectations.

Figure 6 compares the observed monthly number of large declines with the expected counts from both probability models.

The black line represents the observed monthly frequency, while the red and blue lines represent the fitted Poisson and Negative Binomial expected counts.

The observed number of events varies considerably between months. Some months contain no declines exceeding 2%, while the maximum observed frequency reaches six events.

Both fitted models produce relatively stable expected monthly counts, with moderate variations caused by differences in the number of trading days.

The Poisson and Negative Binomial curves are almost identical. This similarity indicates that the more complex Negative Binomial model provides little additional explanatory power for the observed event frequency.

Although individual months deviate from expected values, the observed fluctuations do not by themselves establish that the Poisson model is inadequate. The formal goodness-of-fit analysis is presented in Section 4.

3.4. Probability of an Extreme Daily Loss

To evaluate extreme downside risk, the largest observed one-day loss was selected as the threshold.

The maximum negative logarithmic return corresponded to a loss magnitude of 10.5357%, equivalent to an ordinary percentage price decline of approximately 10.00%.

For each fitted model, the probability of experiencing a loss larger than this historical maximum was estimated.

For the return distributions, the relevant probability is the lower-tail probability:

For the positive loss size models, the exceedance probability is based on:

Since the loss size distributions were fitted only to negative-return days, their tail probabilities were multiplied by the observed proportion of loss days to obtain unconditional probabilities per trading day.

The estimated waiting time was calculated as:

Table 6. Probability of a daily loss exceeding the historical maximum

Model	Probability (%)	Once every N trading days
Normal	0.00000571	17,510,698
Student's t	0.08992	1,112
Skewed Student's t	0.03283	3,046
Gamma	0.00643	15,551
Lognormal	0.74065	135

The estimated extreme-loss probabilities vary substantially across the fitted models.

The Normal distribution produces an extremely small exceedance probability. This reflects its relatively light tails and illustrates why a Normal model may underestimate the probability of severe market movements.

Student's t estimates a substantially larger probability, reflecting the greater frequency of extreme observations permitted by its heavier tails.

The Skewed Student's t model estimates a probability of approximately 0.03283%, corresponding to one event every 3,046 trading days. This estimate reflects the fitted asymmetry and tail behavior of BYD's returns.

The Gamma model predicts a lower probability than Student's t or Skewed Student's t, while the Lognormal model produces the highest exceedance probability among the five models.

In the observed dataset, no daily loss strictly exceeded the historical maximum. One observation was equal to this maximum threshold, corresponding to approximately one occurrence in 968 trading-return observations.

The differences between model estimates demonstrate that extreme-loss probabilities are highly sensitive to distributional assumptions. In particular, models that describe average behavior adequately may provide very different estimates in the far tail.

Therefore, estimated waiting times should be interpreted as model-based long-run averages rather than precise forecasts of future extreme losses.

4. Model Selection and Goodness-of-Fit Tests

4.1. Model Evaluation Methodology

The probability models were evaluated using graphical diagnostics, information criteria, and formal goodness-of-fit tests.

Maximum Likelihood Estimation was used to estimate the model parameters. Model quality was then compared using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).

The AIC is defined as:

The BIC is defined as:

where L is the maximized likelihood, k is the number of estimated model parameters, and n is the sample size.

Lower AIC and BIC values indicate a preferred balance between goodness of fit and model complexity when comparing models fitted to the same variable.

However, information criteria do not establish whether a model adequately reproduces the observed distribution. Therefore, formal goodness-of-fit tests were also performed.

For the continuous distributions of daily returns and loss sizes, the Kolmogorov–Smirnov (KS) statistic was calculated. The KS statistic measures the maximum absolute difference between the empirical and theoretical cumulative distribution functions:

For monthly loss frequencies, a Pearson-type goodness-of-fit statistic was calculated to compare observed event counts with the fitted count models.

Bootstrap simulation with 199 replications was used to estimate p-values, accounting for parameter estimation from the observed data.

The statistical hypotheses were defined as follows:

Null hypothesis (H_0): The observed data are consistent with the selected probability distribution.

Alternative hypothesis (H_1): The observed data are not consistent with the selected probability distribution.

The significance level was set at 0.05.

If the p-value was below 0.05, the null hypothesis was rejected. Otherwise, there was insufficient evidence to reject the null hypothesis.

A failure to reject H_0 does not prove that the model is correct; it indicates that the test did not detect a statistically significant discrepancy at the selected significance level.

4.2. Goodness-of-Fit Analysis of Daily Returns

The Normal, Student's t, and Skewed Student's t distributions were compared using AIC, BIC, the KS test, and Q–Q plots.

Table 7. Goodness-of-fit results for daily return distributions

Model	KS statistic	p-value	AIC	BIC
Normal	0.0610	0.005	−4835.94	−4826.19
Student's t	0.0269	0.015	−4922.51	−4907.89
Skewed Student's t	0.0151	0.565	−4933.70	−4914.20

The Normal distribution produced the highest KS statistic (0.0610) and the least favorable AIC and BIC values among the three models. Its p-value of 0.005 is below the 5% significance level, so the null hypothesis was rejected.

Student's t improved the fit substantially, reducing the KS statistic to 0.0269 and producing a considerably lower AIC. However, its p-value was 0.015, meaning that the null hypothesis was also rejected at the 5% level.

The Skewed Student's t distribution achieved the lowest KS statistic (0.0151), the lowest AIC (−4933.70), and the lowest BIC (−4914.20). Its bootstrap p-value was 0.565, exceeding the significance threshold.

Therefore, the null hypothesis could not be rejected for Skewed Student's t.

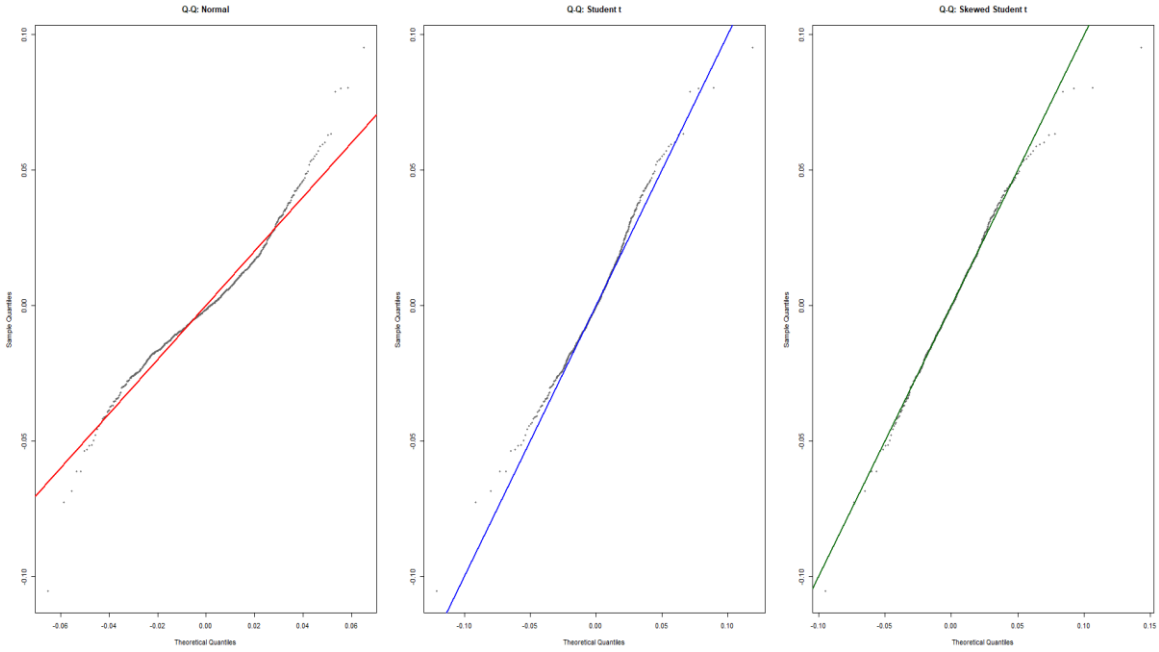


Figure 7. Q–Q plots comparing empirical BYD daily returns with the theoretical quantiles of three fitted distributions.

The Q–Q plots provide further evidence supporting the model comparison.

For the Normal distribution, noticeable deviations from the diagonal occur at both extremes, suggesting that the model underestimates the frequency or magnitude of extreme returns.

The Student's t Q–Q plot shows improved agreement, particularly over the central and moderately extreme ranges. Nevertheless, some observations deviate from the theoretical line.

The Skewed Student's t Q–Q plot shows the closest overall alignment with the reference line, although deviations remain at some extreme observations.

Taken together, the visual diagnostics, information criteria, and KS results support the selection of Skewed Student's t as the most suitable model for BYD's daily logarithmic returns among the three alternatives.

This result indicates that both heavy tails and asymmetry are relevant when describing the distribution of BYD's daily price changes.

4.3. Goodness-of-Fit Analysis of Loss Size

The Gamma and Lognormal models were compared using the same model selection approach.

Table 8. Goodness-of-fit results for loss size distributions

Model	KS statistic	p-value	AIC	BIC
Gamma	0.0445	0.030	1346.09	1354.60
Lognormal	0.1082	0.005	1460.87	1469.38

The Gamma distribution produced a lower AIC and BIC than the Lognormal distribution, indicating better relative performance.

Its KS statistic was also considerably smaller, suggesting that its cumulative distribution function was closer to the empirical distribution.

Nevertheless, the Gamma model produced a p-value of 0.030. Since this value is below 0.05, the null hypothesis was rejected.

The Lognormal distribution performed worse, with a KS statistic of 0.1082 and a p-value of 0.005. Its null hypothesis was also rejected.

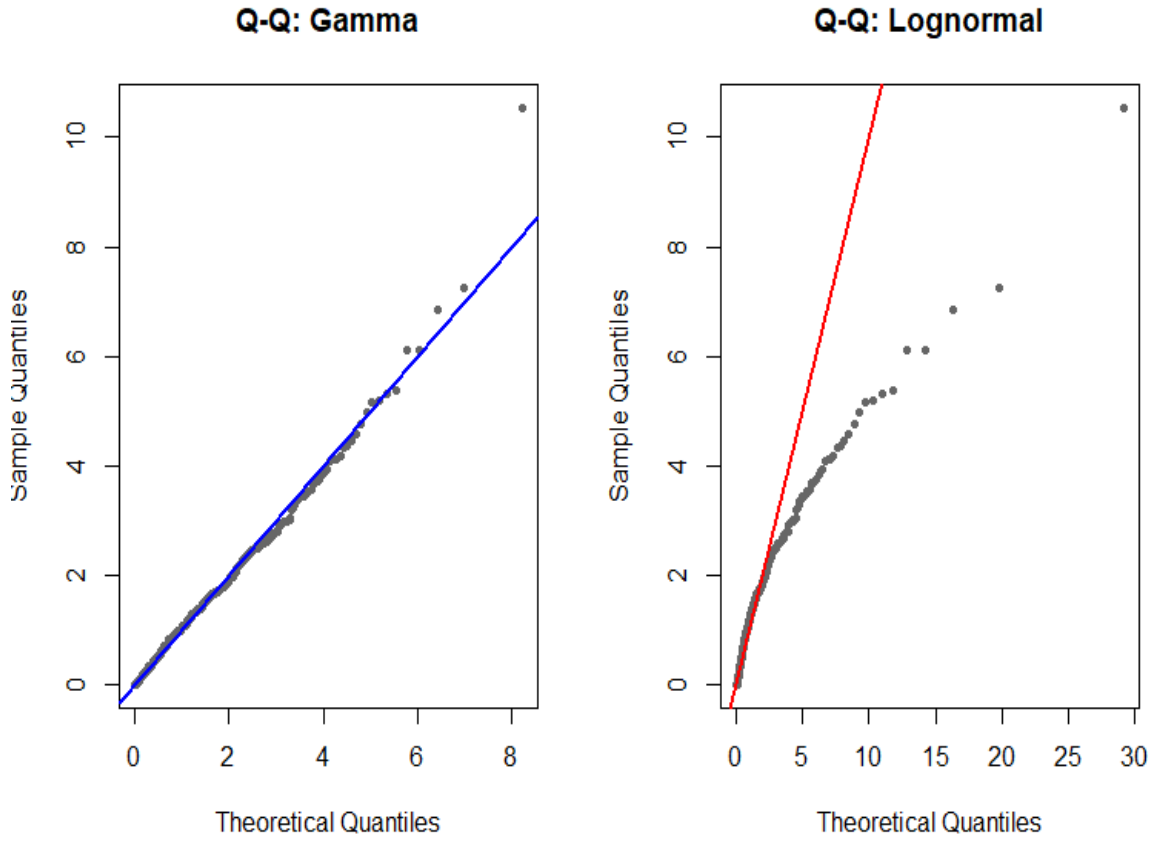


Figure 8. Q–Q plots comparing empirical BYD loss sizes with theoretical Gamma and Lognormal quantiles.

The Gamma Q–Q plot shows that most observations are relatively close to the reference line, especially for small and moderate loss sizes. However, noticeable deviations appear among the largest losses.

The Lognormal Q–Q plot exhibits much stronger deviations, particularly in the right tail. The model predicts theoretical quantiles substantially larger than many of the corresponding observed quantiles.

These graphical results are consistent with the AIC and KS comparisons, which strongly favor the Gamma distribution over the Lognormal distribution.

However, the rejection of both models indicates that neither distribution provides a fully satisfactory description of loss sizes at the 5% significance level.

Thus, Gamma can be selected as the best relative model among the two fitted alternatives, but its adequacy remains limited.

The remaining discrepancies may be associated with unusual extreme observations, differences in market conditions across the sample period, or other characteristics not captured by these relatively simple parametric distributions.

A more flexible distribution or a separate model for extreme losses may improve the description of the loss-size tail. However, such extensions would require additional analysis and statistical validation.

4.4. Goodness-of-Fit Analysis of Monthly Loss Frequency

The monthly number of daily declines exceeding 2% was modeled using Poisson and Negative Binomial distributions.

Since the data represent event counts, a Pearson-type goodness-of-fit statistic was used instead of the standard continuous KS test.

The models accounted for different numbers of trading days in each month through exposure adjustments.

Table 9. Goodness-of-fit results for monthly loss frequencies

Model	Pearson statistic	p-value	AIC	BIC
Poisson	41.317	0.770	167.74	169.63
Negative Binomial	41.305	0.700	169.74	173.53

The Poisson model produced an AIC of 167.74 and a BIC of 169.63, both lower than the corresponding Negative Binomial values.

Its bootstrap p-value was 0.770, indicating insufficient evidence to reject the null hypothesis.

The Negative Binomial model also passed the goodness-of-fit test, with a p-value of 0.700. However, its additional dispersion parameter did not produce an improvement sufficient to compensate for the increased model complexity.

Furthermore, the estimated Negative Binomial dispersion parameter approached a very large value, causing the model to behave similarly to a Poisson distribution.

This behavior explains why the fitted curves almost completely overlap in Figure 6.

Therefore, the Poisson model was selected as the preferred distribution for monthly loss frequency. It provides an adequate fit according to the applied test while using fewer parameters.

The result suggests that the observed monthly frequency of declines exceeding 2% does not show sufficient additional dispersion to justify the more complex Negative Binomial model within the fitted specification.

4.5. Final Model Comparison

The results of the statistical tests and information criteria were combined to identify the most suitable probability model for each variable.

Table 10. Final model selection

Variable	Selected model	AIC	p-value	Conclusion
Daily Returns	Skewed Student's t	−4933.70	0.565	Not rejected
Loss Size	Gamma	1346.09	0.030	Best AIC; rejected
Loss Frequency	Poisson	167.74	0.770	Not rejected

The Skewed Student's t distribution provided the strongest overall performance for daily returns. It achieved the best AIC and BIC values and was not rejected by the goodness-of-fit test.

The Gamma distribution was preferred for modeling positive daily losses because it outperformed the Lognormal model in the comparison of AIC, BIC, empirical moments, and Q–Q plots. However, its statistically significant goodness-of-fit result indicates that it should not be treated as an entirely adequate description of the observed loss sizes.

For monthly loss frequency, the Poisson model was selected because it provided an acceptable goodness-of-fit result and was more parsimonious than the Negative Binomial alternative.

Overall, the results demonstrate that no single probability distribution is suitable for every aspect of BYD's stock risk. Daily returns require a flexible model capable of representing heavy tails and asymmetry, while loss magnitudes require a separate positive-valued severity distribution and loss frequency requires a discrete count model.

The selected models provide a structured framework for distinguishing between the magnitude and frequency of losses. At the same time, the inadequate fit of both considered severity models emphasizes the importance of validating assumptions and acknowledging uncertainty, especially when estimating extreme financial losses.

5. Final Conclusion

This study examined the financial returns and associated investment risks of BYD Company Limited (002594.SZ) using historical stock market data from October 19, 2022, to October 19, 2026. The analysis included 969 adjusted daily closing prices and 968 daily logarithmic returns. The main objective was to investigate the statistical characteristics of stock returns, identify the frequency and magnitude of losses, and determine which probability distributions provide the most appropriate representation of the observed risks.

The exploratory data analysis revealed that BYD's stock price experienced considerable fluctuations throughout the observation period. After declining in early 2024, the share price increased substantially during 2024–2025 before falling again. Despite these significant movements, the average daily logarithmic return remained close to zero (−0,00003), while the standard deviation was 0,01987, equivalent to approximately 1,99 percentage points. The positive skewness of 0,356 and kurtosis of 5,501 indicated that the distribution was asymmetric and characterized by heavier tails than the Normal distribution. These findings demonstrate that relatively small average returns can coexist with substantial short-term investment risk.

The analysis of individual losses provided further evidence of this risk. Negative returns were recorded on 520 out of 968 trading days, representing 53,72% of the observations. The largest one-day decline occurred on April 7, 2025, when BYD shares lost 10,00% of their adjusted value. Other significant declines coincided with periods of economic uncertainty, changes in market expectations, international trade tensions, and competitive pressure in the electric vehicle industry. These observations suggest that BYD's stock performance is sensitive to both broader market developments and industry-specific conditions.

Three probability distributions were examined for daily logarithmic returns: Normal, Student's t, and Skewed Student's t. Based on the information criteria, graphical diagnostics, and goodness-of-fit tests, **Skewed Student's t was identified as the most suitable distribution**. It achieved the lowest AIC (−4933,701) and BIC (−4914,200), together with a bootstrap KS test p-value of 0,565. Therefore, the null hypothesis was not rejected at the 5% significance level. In

contrast, the Normal and symmetric Student's t distributions were rejected by the corresponding tests. The superior performance of Skewed Student's t indicates that accounting for both asymmetry and heavy tails is important when modeling BYD's daily stock returns.

For the distribution of positive loss magnitudes, the Gamma and Lognormal models were compared. The Gamma distribution demonstrated better relative performance, achieving a substantially lower AIC (1346,095) than the Lognormal distribution (1460,871). Its theoretical mean and variance were also closer to the corresponding empirical statistics. Nevertheless, the Gamma model produced a goodness-of-fit p-value of 0,030, while the Lognormal model produced a p-value of 0,005. As both values were below the 5% significance threshold, both models were statistically rejected. Consequently, Gamma can be considered the better of the two examined alternatives, but neither provides a fully satisfactory representation of the observed loss severity. This limitation is especially relevant when estimating the probabilities of unusually large losses.

The monthly frequency of daily stock price declines exceeding 2% was analyzed using Poisson and Negative Binomial distributions. The **Poisson distribution was selected as the preferred frequency model**, with an AIC of 167,743 and a bootstrap goodness-of-fit p-value of 0,770. The Negative Binomial model did not provide a meaningful improvement and converged toward the Poisson limit. This suggests that, after accounting for differences in the number of trading days per month, the observed variation in substantial loss frequency did not require an additional overdispersion parameter within the considered models.

The extreme-loss analysis further demonstrated the sensitivity of financial risk estimates to distributional assumptions. The largest observed negative logarithmic return was approximately −10,54%, corresponding to a simple daily price decline of 10,00%. The fitted Skewed Student's t distribution estimated the probability of a more severe daily loss at approximately 0,03283%, equivalent to an expected occurrence of once every 3 046 trading days. However, the estimated probabilities differed considerably across the examined models. This variation demonstrates that extreme-risk estimates depend strongly on the assumed probability distribution and should be interpreted with caution rather than treated as precise predictions.

Overall, the research shows that **no single probability distribution is sufficient to describe every aspect of BYD's investment risk**. Skewed Student's t provides the strongest statistical description of daily returns, Gamma offers the best relative approximation of loss severity despite failing the goodness-of-fit test, and Poisson adequately describes the monthly frequency of substantial declines. These findings emphasize the importance of examining return distributions, loss magnitudes, and loss frequencies separately when conducting financial risk assessments.

The study also has several limitations. The analysis is based on historical observations and assumes that fitted statistical relationships are informative about future market behavior. However, stock market conditions, regulatory policies, competition, and macroeconomic factors may change over time. Furthermore, the models do not explicitly account for volatility clustering, potential structural changes, or dependence between daily returns. Future research could incorporate time-varying volatility models, more flexible severity distributions, and methods specifically designed for extreme financial losses.

In conclusion, BYD shares exhibited substantial price variability and exposure to extreme daily movements despite an average return close to zero. The results confirm that the selection and validation of probability distributions are essential for meaningful investment risk assessment.

Although the fitted models provide useful information about historical patterns, their limitations must be recognized when evaluating potential future losses.